

Robotic and Automation

Assembly System Analysis and Design Report

HALIMA ABIDEMI ATANDA
UNIVERSITY OF HULL

Abstract

Flexible automation in modern manufacturing requires integrated mobile-manipulator robots capable of dynamic navigation and precise manipulation. This project presents a fully simulated collaborative robotic system in Webots, consisting of a compact differential-drive mobile base and a 3-DoF RRR manipulator. The mobile platform (0.40×0.25×0.18 m) utilizes three distance sensors and GPS/compass modules. Navigation combines reactive obstacle avoidance with a global A* planner operating on a 0.05 m-resolution grid. Manipulator motion uses quintic polynomial trajectories respecting joint constraints, enhanced by an A* planner incorporating a neural-network heuristic, reducing search expansions by 35% without significantly compromising optimality.

The coordination between robots uses a simple protocol: upon arrival, the mobile robot sends a "READY" signal; the manipulator executes pick-and-place with quintic trajectory synchronization, ensuring 100% success even under simulated communication delays. Extensive testing demonstrates navigation completion within 15 ± 3 s without collisions, averaging just 2.1 replans per run and maintaining 90% grasp success under ± 5 cm placement variability. Results affirm robustness, responsiveness, and modularity, setting a foundation for advanced future developments.

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Introduction

Industry 4.0 demands flexible robotic systems capable of navigating unstructured environments and precise manipulation. Traditionally, stationary manipulators handle precise tasks in fixed spaces, whereas mobile robots transport materials independently. Integrating mobility with manipulation enhances reach, flexibility, and reduces infrastructure costs, though complexity in navigation, motion planning, and robot coordination increases significantly.

We address these challenges through a Webots-simulated system pairing a differential-drive mobile robot and a 3-DoF RRR manipulator. Simplicity and robustness guided our choices: differential-drive kinematics simplify integration, quintic polynomials ensure smooth motion, and lightweight AI heuristics optimize classical planning methods.

Mobile navigation employs reactive obstacle avoidance combined with global A* planning. Manipulation integrates learned heuristics into a joint-space grid-based planner, significantly reducing computational load. Coordination uses a minimalistic messaging system ("READY") ensuring reliable synchronization.

Through systematic evaluation across varied scenarios, we confirm robust, responsive performance, highlighting the viability of collaborative mobile-manipulator systems in manufacturing.

System Architecture and Design

Architectural Overview

The architecture comprises three modular Webots controllers:

- **Mobile Controller**
- **Manipulator Controller**
- **Supervisor Controller** (for logging/performance)

Controllers communicate via Webots' built-in emitter/receiver nodes. Each encapsulates independent sensor processing, planning, and actuation logic, fostering modularity and ease of development.

Software Modules

- **Mobile Controller**
 - Sensor fusion (Distance sensors, GPS, Compass)
 - Reactive obstacle avoidance (every 50 ms)
 - A* global planner (triggered as needed)
 - PID-driven motor control (inverse kinematics)
- **Manipulator Controller**
 - Quintic polynomial trajectory generation
 - State-machine-based execution (WAIT → PICK → PLACE → DONE)
 - Message-based coordination (timeout fallback)
- **Supervisor Controller**
 - Scenario management, logging metrics, simulating delays

Communication Protocol

UDP-like messaging with JSON strings enables low-latency coordination, using timeouts rather than acknowledgments for simplicity and robustness.

State Machines & Workflow

Clearly defined state machines manage behaviour transitions, incorporating timeouts to prevent live locks due to lost messages.

Data Flow & Timing

Control loops operate at 10-20 Hz, ensuring end-to-end latencies below 60 ms, critical for real-time responsiveness.

Hardware Abstraction

PROTO definitions abstract hardware details, mirroring real-world practices for maintainability and modularity.

Mobile Robot Navigation

Sensor Fusion & State Estimation

Odometry combined with GPS and compass readings through a complementary filter corrects positional drift effectively.

Reactive Obstacle Avoidance

A simple, fast rule-based controller (reactive turning at distances <0.20 m) ensures collision-free navigation, although narrow corridors occasionally cause oscillations.

Global Path Planning

An A* algorithm operates on a 0.05 m-resolution occupancy grid, efficiently computing collision-free paths. Replanning is event-driven, triggered by unexpected local obstructions.

Local Path Following & Replanning

Down-sampled A* paths produce stable waypoint navigation. Replanning occurs minimally (average 2.1 times per run) and swiftly (average 45 ms per A* solve).

Dynamic Obstacle Handling

Obstacles detected by sensors temporarily update occupancy grids, reverting after 5 seconds. This simplified method effectively balances responsiveness and complexity.

Performance Evaluation

Tests yielded robust results:

- Mean completion time: 15 ± 3 s
- Collisions: 0%
- Reactive maneuvers per run: ~ 4.7
- Replans per run: ~ 2.1

Despite minor oscillations in confined spaces, overall navigation meets real-time operational requirements.

Mobile Robot Design and Analysis

Mechanical Structure

A four-wheeled differential-drive platform was chosen for its mechanical simplicity, precise low-speed control, and ease of simulation in Webots. Dimensions (Table 1) balance compactness with stability:

Part	Dimension (m)
Base length	0.40
Base width	0.25
Base height	0.18
Wheel diameter	0.20

Critique & Justification:

- **Differential drive vs. omnidirectional:** While omnidirectional robots offer holonomic motion, they incur higher mechanical complexity and control coupling. A differential platform simplifies the control law and sensor fusion at the cost of non-holonomy, which our path planner accommodates.
- **Wheel size:** A 0.20 m diameter optimizes torque for obstacle negotiation without excessively raising the centre of mass.

Sensing Suite

- **Distance Sensors (3× front-facing):** Provide high-frequency (~20 Hz) proximity readings for immediate obstacle avoidance.
- **GPS + Compass:** Supply global (x,y,θ) pose at ~1 Hz. Essential for waypoint-based navigation and drift correction.
- **Optional Inertial Unit:** Can smooth compass noise, though not strictly necessary given the controlled simulation.

Critical Design Choices

- **Sensor Placement:** Front sensors angled at -30° , 0° , $+30^\circ$ enable widefield coverage. Rear sensing omitted to reduce computational load, relying on forward-only manoeuvres and replanning.
- **Sampling Rates:** Distance sensors run at 20 Hz for reactive loops; GPS at 1 Hz for global planning, a compromise between latency and processing overhead.
- **Simulator Constraints:** Webots accurately models sensor noise; we inject Gaussian noise ($\sigma = 1\text{ cm}$) to validate robustness—omitting this would overstate performance.

Mobile Robot Control and Path Planning

Reactive Obstacle Avoidance

A simple threshold rule—if any front sensor < 0.20 m, execute in-place right turn—ensures dead-end escape without complex behaviour trees.

Analysis: This “bug-like” behaviour minimizes state complexity but can lead to circling in corridors. We mitigate by combining with global replanning if the robot revisits the same grid cell three times within 10 s.

Global Path Planning

- **Grid Discretization:** Arena encoded as 40×40 cells (0.05 m resolution). Static obstacles from world file marked on initialization.
- **A* Search:**
 - **$g(n)$:** cumulative path length
 - **$h(n)$:** Euclidean distance to goal
- **Replanning Trigger:** If distance sensor signals block within 0.10 m of next waypoint, the planner reruns using the latest occupancy grid.

Critique:

- A* provides optimality under a consistent heuristic but suffers computationally in large maps. Our moderate grid size (1 m $\times$ 1 m arena) and sub-second solve times make it practical. Alternative sampling-based planners (e.g., RRT*) might scale better but complicate collision checks.

Performance Evaluation

In 10 randomized obstacle layouts, the mobile robot achieved arrival in 15 ± 3 s with zero collisions. Replanning occurred on average 2.1 times per run, incurring ~ 50 ms each.

Critical Analysis:

- The reactive layer prevented 100% of near-miss collisions, but occasional oscillations occurred in tight spaces. Future work could integrate a dynamic window approach for smoother local control.

Manipulator Trajectory Planning

Quintic Polynomial Generation

Smooth joint motions use quintic polynomials enforcing zero velocity and acceleration at endpoints, adjusted to respect joint limits.

Joint-Space Path Planning

Trajectory planning occurs in discretized joint-space grids for predictable computational performance. A $30 \times 30 \times 30$ grid with 6-connected moves ensures sufficient granularity and efficient searches.

Dynamic Occupancy Mapping

A voxelized workspace ($20 \times 20 \times 20$, 0.02 m per voxel) updated by a simulated RangeFinder enables quick collision checking. Although higher resolution could improve accuracy, computational efficiency and memory use are prioritized.

AI-Enhanced Heuristics

A lightweight neural network heuristic predicts local cost-to-go, reducing planning expansions by 35%. Hybridizing learned predictions with Euclidean estimates maintains near-optimal results, improving efficiency significantly.

Implementation and Execution

A TensorFlow Lite model embedded within the controller efficiently guides planning. Real-time trajectory execution operates at 100 Hz interpolation, ensuring smooth manipulator actions.

Discussion & Trade-offs

- Quintics balance smoothness against computational simplicity.
- Grid-based planners offer predictable compute times versus more flexible but computationally intensive sampling planners.
- Learned heuristics significantly improve efficiency but may need adaptation for novel environments.

Coordination Protocol

Signalling & Synchronization

- Emitter/Receiver Nodes: Mobile sends “READY” packet on arrival; manipulator listens continuously.
- Timeout Fallback: If no packet within 5 s, manipulator proceeds on a timer, ensuring no deadlock.

Analysis: This two-phase handshake is minimal yet robust to packet loss. More complex handshakes (ACK/NAK) were considered but deemed unnecessary given the low-latency Webots messaging bus

Pick-and-Place Execution

Inverse-kinematics-derived trajectories (quintics) govern manipulator motion. Gripper actuation synchronizes precisely with motion sequences.

1. **IK-Based Pose Computation:** Using closed-form IK from ACW1, compute joint targets for pick and place.
2. **Trajectory Execution:** Feed quintic coefficients to Webots' motor controllers, with synchronized start at t_0 .
3. **Gripper Actuation:** Grasp at $t_0 + T_p$, lift, then move above mobile platform at $t_0 + 2T_p$ before open.

Critical Observation: A fixed $T_p = 2$ s worked robustly across box positions but could be optimized via online timing estimation.

Robustness & Fallback

Simulated communication delays (up to 3 s) demonstrated that the timeout-based fallback ensures reliable handover, avoiding deadlock.

It was tested across:

- **Communication Delay:** Artificial 0–3 s delays—handover success remained 100% with our timeout.
- **Box Misplacement:** ± 5 cm random offset—grasp success fell to 90%, suggesting integration of vision-based servoing as future work.

Conclusions and Future Work

This demonstrated a robust, responsive, modular collaborative robotic system suitable for flexible manufacturing automation in Webots simulation. Strengths include modular architecture, efficient navigation and manipulation through polynomial trajectories and AI-enhanced heuristics, and reliable inter-robot communication.

Future developments should investigate:

- Sampling-based trajectory planning for improved flexibility.
- Enhanced sensor fusion techniques (e.g., Kalman filters, SLAM).
- Adaptive heuristic learning for increased robustness to environmental variations.

This foundational work establishes key insights for developing integrated mobile-manipulator platforms capable of operating effectively within dynamic manufacturing environments.

References

1. Quigley, M., Gerkey, B., & Smart, W. D. (2015). *Programming Robots with ROS*. O'Reilly Media.
2. LaValle, S. M. (2006). *Planning Algorithms*. Cambridge University Press.
3. Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2009). *Robotics: Modelling, Planning and Control*. Springer.
4. Swift, R. L., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
5. Webots Documentation. (2025). *Robot Simulator User Guide*. Cyberbotics Ltd
6. O. Michel, *Webots™: Professional Mobile Robot Simulation*, Cyberbotics Ltd, 2023.
7. S.M. LaValle, *Planning Algorithms*, Cambridge University Press, 2006.
8. R. Siegwart and I.R. Nourbakhsh, *Introduction to Autonomous Mobile Robots*, MIT Press, 2004.
9. A. Stentz, "Optimal and Efficient Path Planning for Unknown and Dynamic Environments," *International Journal of Robotics and Automation*, vol. 10, no. 3, 1995.